

Tecnun
Universidad
de Navarra

Analysis of breast cancer patients with both supervised and unsupervised learning

Carlos Peter Tesoriero Garduño
Felipe Salazar Molina

Machine Learning

1: Introduction.....2
1.1 Background of Machine Learning in medical diagnosis.....3
1.2 Data Set.....3
1.3 Objectives.....3
1.4 Methodology.....4
2: Data overview, quality, and exploratory data analysis.....4
2.1 Loading and target encoding.....4
2.2 Shape and data quality.....5
2.3 Train/test split.....5
2.4 Exploratory data analysis on the training set.....6
2.5 Feature scaling.....10
2.6 Preprocessing strategy.....10
3: Unsupervised learning.....11
3.1 Principal Component Analysis.....11
3.2 t-SNE.....12
3.3 Clustering.....13
3.4 What we learned from unsupervised learning.....16
4: Supervised learning.....16
4.1 Setup.....16
4.2 Models on the original 30 features.....16
4.3 Models with PCA inside the pipeline.....17
4.4 Model selection.....18
5: Test-set evaluation.....18
5.1 Final metrics.....18
5.2 Classification report.....18
5.3 Confusion matrix.....18
5.4 Medical interpretation.....19
6: Connecting unsupervised and supervised learning.....20
6.1 PCA improves prediction marginally but consistently.....20
6.2 The clusters foreshadow the classifier.....20
6.3 Same axis throughout.....20
6.4 Possible extensions.....20
7: Reflection.....21
8: Conclusion.....21

1: Introduction

1.1 Background of Machine Learning in medical diagnosis

Reviewing thousands of patient records to find a common pattern behind a certain disease becomes near impossible for humans to do by hand. Handling that volume of information, and then being able to separate it into readable and predictable results, is not viable manually.

With the rise of machine learning, that task can be given to an algorithm, which extracts features from thousands of data points and learns the patterns that distinguish one outcome from another. This is extremely useful in cancer diagnosis, as it can be used to detect early signs from previous patient data, track tumor growth, analyze the stage of the disease, and so on.

There are two main approaches to this kind of learning. In supervised learning, the model is trained on labeled data and learns to map inputs to known outcomes through tuned parameters. In unsupervised learning, no labels are used; the algorithm instead looks for structure in the data on its own, usually through dimensionality reduction or clustering.

1.2 Data Set

The dataset chosen for this project is the Breast Cancer Wisconsin (Diagnostic) dataset, loaded directly from scikit-learn through: `sklearn.datasets.load_breast_cancer()`. It contains 569 patient samples, each described by 30 numerical features extracted from digitized images of fine-needle aspirate (FNA) cell nuclei. Each sample is labeled as either benign or malignant.

For this project, we re-encoded the target so that malignant = 1 and benign = 0. This makes malignant the positive class, which is the medically critical one to detect, and it makes metrics like precision, recall, and F1 directly interpretable for the question we actually care about: how often the model catches a real malignant case.

From this dataset we ask two parallel questions. On the unsupervised side, we ask whether the geometry of the 30 features already reflects the benign/malignant distinction without ever using the labels. On the supervised side, we ask whether a classifier can accurately predict the diagnosis, and whether compressing the features with PCA improves that prediction.

1.3 Objectives

The objectives of this project are:

- To explore the Breast Cancer Wisconsin dataset through unsupervised methods (PCA, t-SNE, K-Means, and Gaussian Mixture Models) and check whether the natural structure of the data reflects the diagnostic labels.
- To build and compare supervised classifiers (Logistic Regression, Support Vector Machine with an RBF kernel, and K-Nearest Neighbors) for predicting malignant versus benign tumors.
- To test whether dimensionality reduction with PCA improves predictive performance, by running every supervised model both on the original 30 features and on PCA components inside the pipeline.
- To evaluate model performance using metrics suited to a medical context (F1, recall, precision, ROC AUC) and to draw conclusions about which model is most convincing.
- To connect the unsupervised and supervised parts of the project, and to see what worked, what did not, and reflect on it.

1.4 Methodology

The project uses a combination of unsupervised and supervised pipeline in Python. The dataset is first loaded and inspected for missing values, duplicates, and feature distributions. It is then split into training (80%) and test (20%) subsets. We only conduct preprocessing on the training sub set of data in order to do an accurate unsupervised exploration. This avoids any data leakage between training and test sets.

In the unsupervised phase, PCA, t-SNE, K-Means, and Gaussian Mixture Models are applied to the standardized training set, and the diagnosis labels are only used afterwards for interpretation. In the supervised phase, Logistic Regression, SVM with an RBF kernel, and K-Nearest Neighbors are tuned through stratified 5-fold cross-validation on the training set, using F1 as the main selection metric. Each model is trained twice, once on the original 30 features and once with PCA inside the pipeline. The single best model is then evaluated once on the untouched test set, with a focus on recall and the confusion matrix because of the medical context.

2: Data overview, quality, and exploratory data analysis

2.1 Loading and target encoding

The dataset is loaded directly from scikit-learn. The features are placed into a DataFrame, while the original target is stored separately. By default, scikit-learn encodes the target as 0 for malignant and 1 for benign, which is the opposite of what we want for this project. Since malignant is the medically critical class, we re-encode the target so that malignant = 1 and benign = 0. This way malignant becomes the positive class, and metrics such as precision, recall, and F1 directly answer the question we actually care about: how well the model identifies real cancer cases.

```
y = y_original.map({0: 1, 1: 0})  # malignant = 1, benign = 0
```

We then combine the features and the relabeled target into a single DataFrame for inspection, with an additional diagnosis_label column that maps the numerical target back to a readable string for plots.

2.2 Shape and data quality

The dataset contains 569 samples described by 30 numerical features, all of type float64. There are no missing values and no duplicated rows, which is unusual for a real-world dataset but reasonable here because the data is curated and was collected for research purposes. As a result, no imputation, encoding, or row-removal step is needed at this stage. The cleaned feature matrix and target vector are therefore identical to the originals.

2.3 Train/test split

Before any analysis, the data is split into a training set (80%) and a held-out test set (20%) using stratified sampling. Stratification ensures that the proportion of benign and malignant samples is similar in both subsets, which matters here because the classes are not perfectly balanced.

```
X_train, X_test, y_train, y_test = train_test_split(
    X_clean, y_clean,
    test_size=0.2,
    random_state=RANDOM_STATE,
    stratify=y_clean
)
```

After the split, the training set contains 455 samples and the test set contains 114 samples. The class proportions are preserved on both sides. From this point on, the test set is not touched until the final evaluation.

2.4 Exploratory data analysis on the training set

The exploratory analysis is performed only on the training set so that any decisions about features, scaling, or modelling are not influenced by the test data.
Class balance. The training set contains roughly 62% benign and 38% malignant samples. The classes are not perfectly balanced, but both are well represented. This confirms why stratifying the split was a useful precaution.

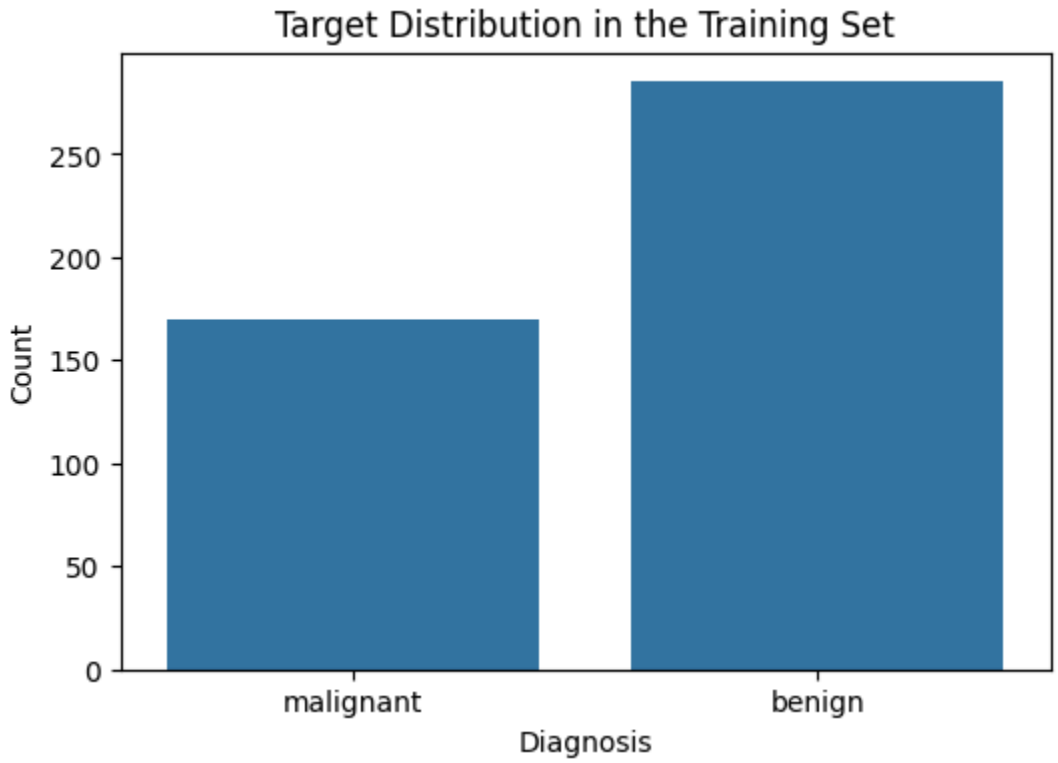


Figure 1: Class distribution in the training set

Feature distributions.
A subset of six representative features is selected for visual inspection: mean radius, mean texture, mean perimeter, mean area, mean concavity, and worst concave points. The histograms show that the features span very different numerical scales: area-related features take values in the hundreds or thousands, while concavity-related features sit on a scale closer to zero. Several features are also right-skewed and have a few extreme values, which is expected for biological measurements.

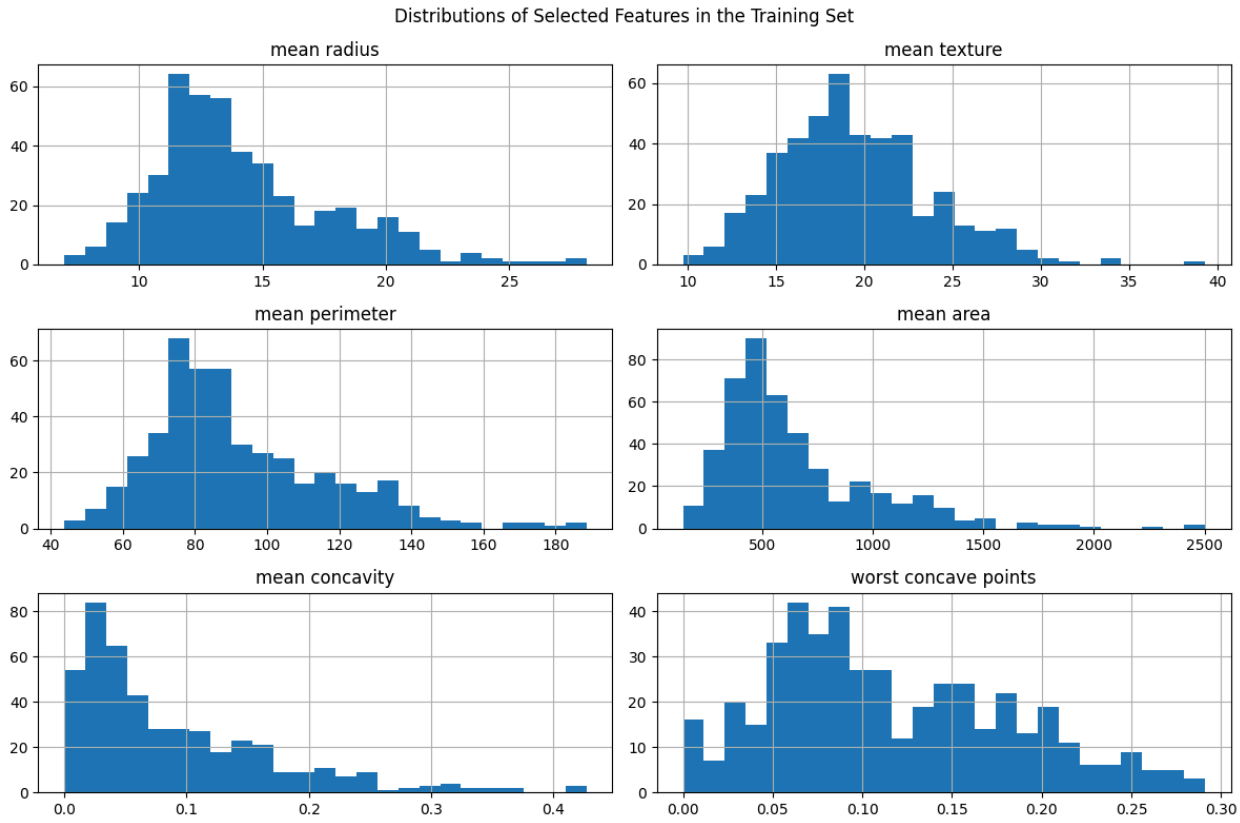


Figure 2: Distributions of selected features

The class-conditional boxplots show that malignant tumors generally have larger and more irregular nuclei, with higher values for radius, perimeter, area, concavity, and concave points. This is a useful sanity check, since these are exactly the kinds of variables that pathologists pay attention to when making a manual diagnosis.

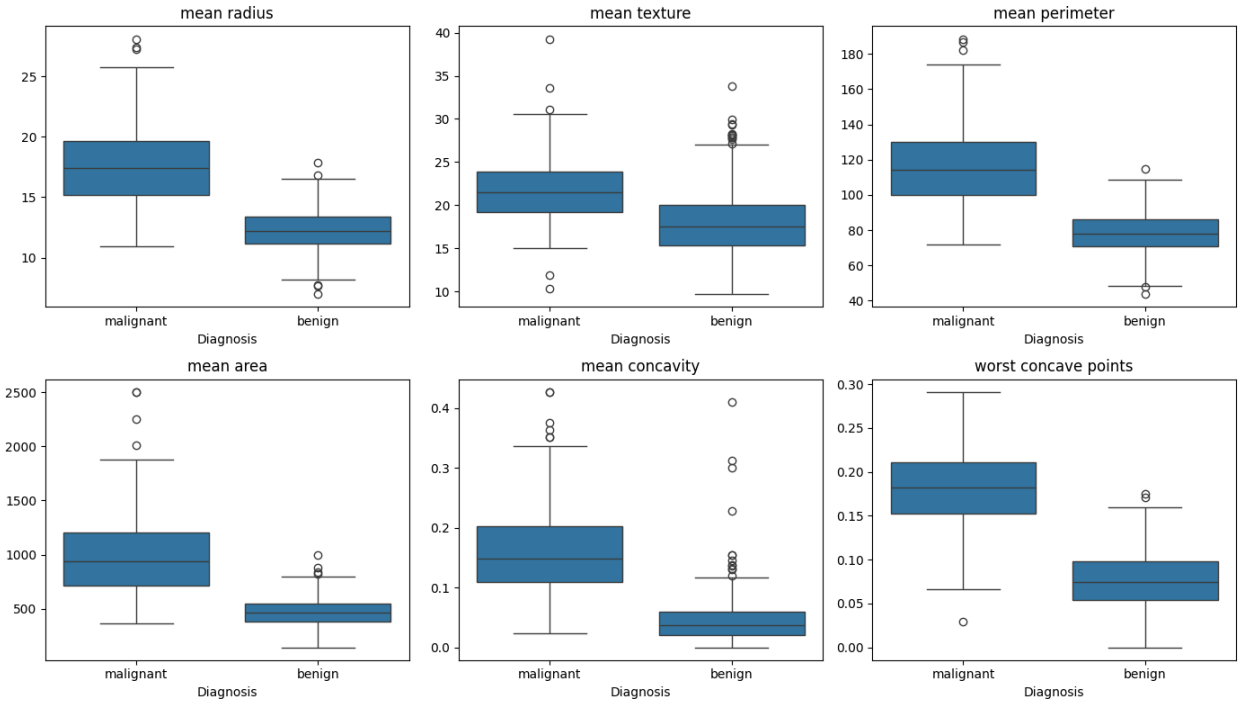


Figure 3: Boxplots of selected features by diagnosis

Feature-feature correlation.

The correlation matrix reveals that many features carry redundant information. Radius, perimeter, and area are almost perfectly correlated, which is geometrically intuitive: they are different ways of measuring the same underlying size. Several worst-case and standard-error variants of the same feature are also strongly correlated with their mean counterparts. Rather than removing these features manually, we keep the full feature set and let PCA later compress this redundancy in a principled way.

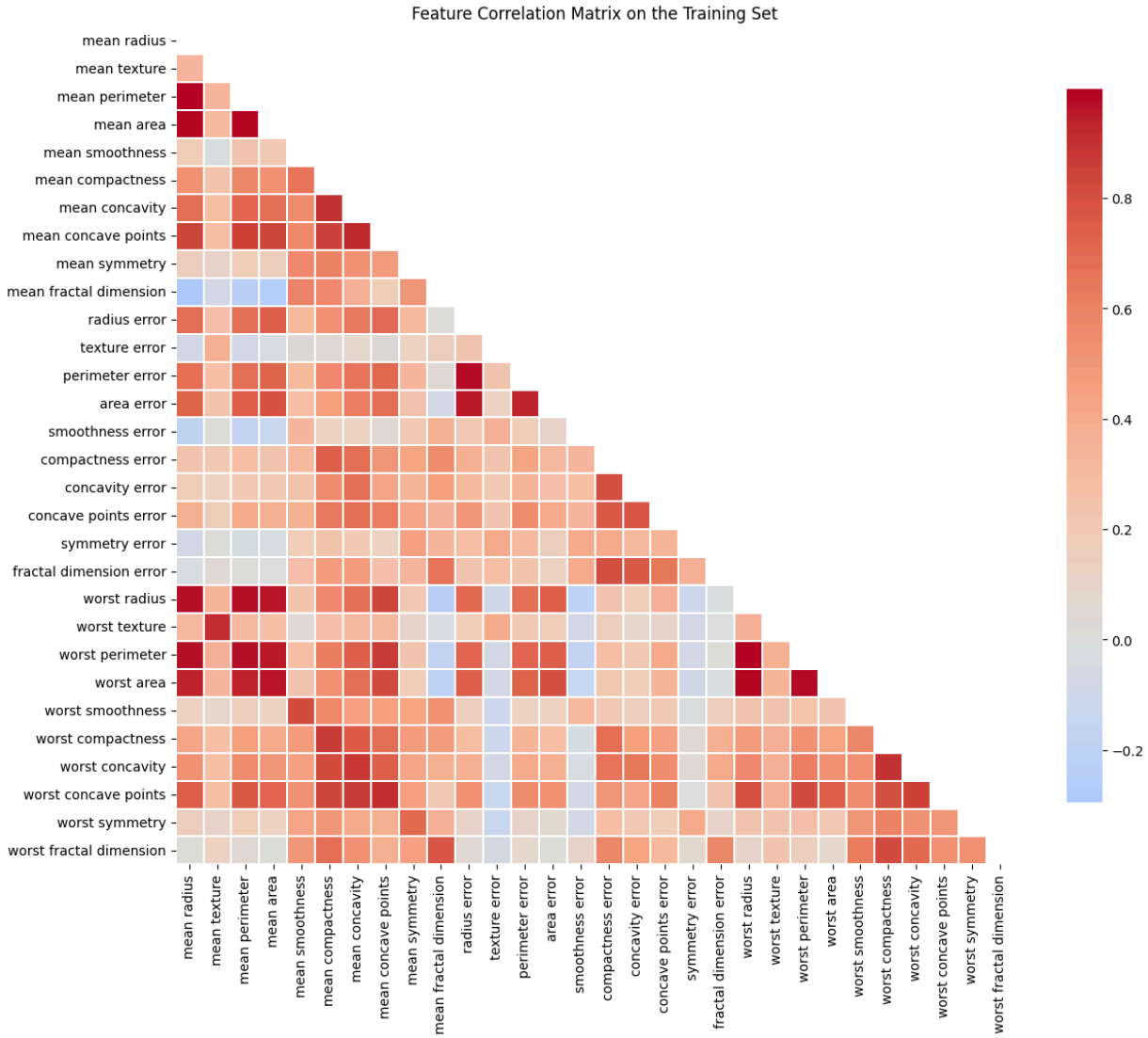


Figure 4: Feature correlation matrix

Feature-target correlation.

Ranking the features by their absolute correlation with the diagnosis, the strongest predictors are concave-points, perimeter, radius, and area variables. This agrees with the boxplots and gives an early indication that the diagnostic signal is concentrated in a small number of related feature directions.

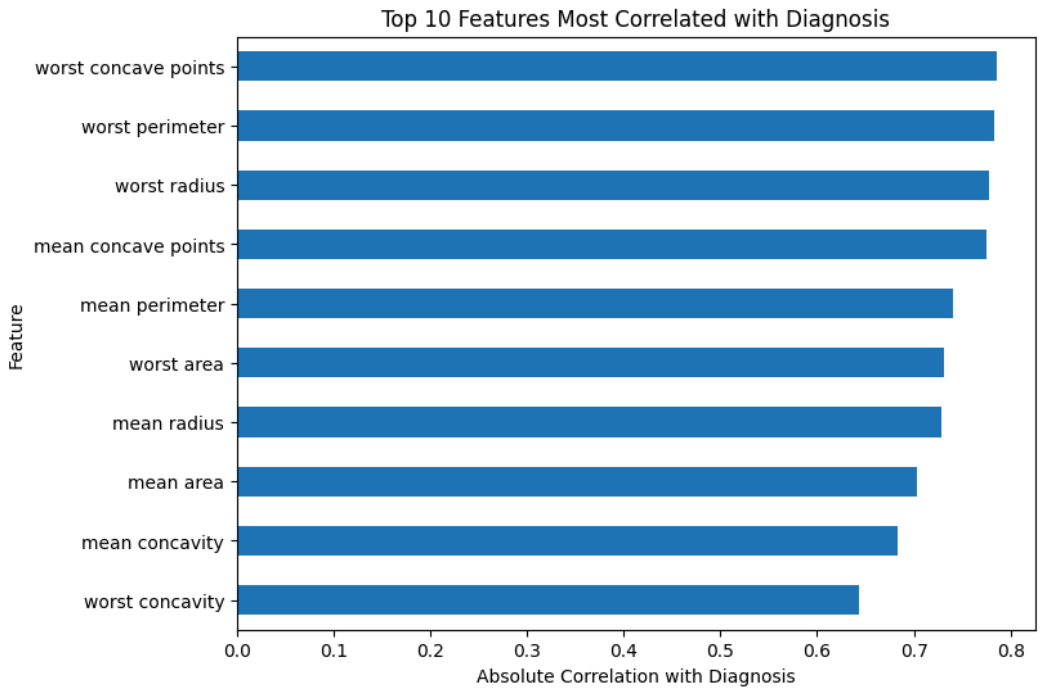


Figure 5: Top features by absolute correlation with diagnosis

---

2.5 Feature scaling

The summary statistics confirm that the features have very different magnitudes. The mean and standard deviation of geometric/size variables are several orders of magnitude larger than those of smoothness or symmetry variables. This matters because most of the methods used later (PCA, t-SNE, K-Means, K-Nearest Neighbors, Support Vector Machines, and Logistic Regression with regularization) are sensitive to feature scale. Without standardization, variables with larger numerical ranges would dominate variance-based, distance-based, or margin-based decisions.

To address this, we apply scikit-learn's StandardScaler. After scaling, every feature has approximately zero mean and unit standard deviation on the training set.

2.6 Preprocessing strategy

The preprocessing strategy depends on the type of method. For the unsupervised analysis (PCA, t-SNE, K-Means, and GMM), a single StandardScaler is fitted only on the training set and then used to transform both training and test data. For the supervised models, scaling is included inside an sklearn Pipeline. This way, during cross-validation, each fold fits its own scaler (and PCA, where applicable) on the training portion of the fold only, which prevents leakage from the validation portion.

This separation keeps the analysis clean and makes sure that any later performance number is honest with respect to unseen data.

3: Unsupervised learning

The first half of the analysis focuses on unsupervised methods. The aim is not to use the diagnosis labels for modelling, but to check whether the structure of the input features already reflects the benign/malignant distinction. The diagnosis is only brought back at the end of each step as a way of interpreting the discovered structure.

3.1 Principal Component Analysis

Principal Component Analysis (PCA) projects the standardized 30-dimensional data onto a lower-dimensional space that preserves as much variance as possible. The first two principal components are used for visualization, while the full PCA is then used to study how the explained variance accumulates with more components.

Two-component projection. Fitting PCA with two components on the scaled training data gives an explained variance ratio of 44.6% for PC1 and 18.5% for PC2. Together, the first two components capture about 63.1% of the total variance, which is a substantial compression given that the data originally lived in 30 dimensions.

When the PCA projection is plotted with the diagnosis used only as a colour, a clear partial separation emerges. Benign samples concentrate on one side of the plot, malignant samples on the other, and the separation runs mostly along the first principal component. Because PCA never saw the labels, this means the diagnostic distinction is already embedded in the geometry of the input features. The separation is not perfect, however; there is overlap in the central region, which suggests that two components alone are not enough for a clean prediction.

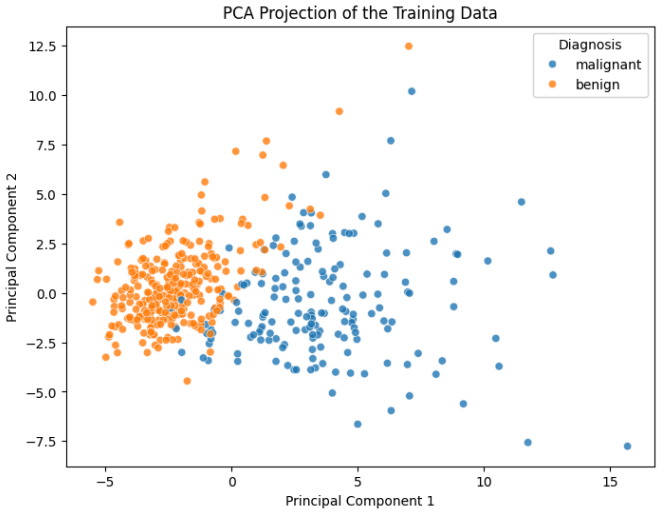


Figure 6: PCA projection of the training data, coloured by diagnosis

Cumulative explained variance.

Fitting PCA with all components and computing the cumulative explained variance answers a slightly different question: how many components are needed to retain most of the information in the original data? The answer is 7 components for at least 90% of the variance and 10 components for at least 95%.

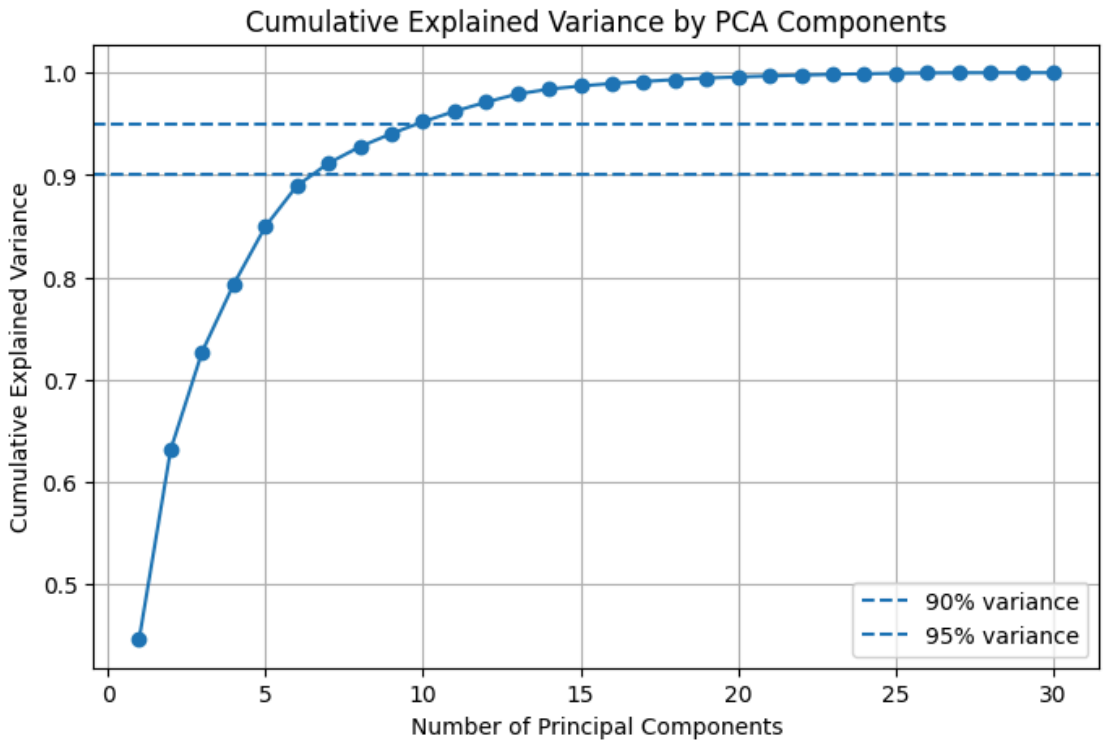


Figure 7: Cumulative explained variance by PCA components

This is useful because it confirms that the 30 features don't really add a lot of information to the differentiation. It also motivates the later supervised experiments, where PCA is inside the pipeline to see whether this redundancy hurts predictive performance.

3.2 t-SNE

t-SNE is then applied as a nonlinear alternative to PCA. While PCA focuses on directions of maximum variance, t-SNE focuses on preserving local neighborhoods, which often reveals structure that linear projections miss.

Because t-SNE is sensitive to its perplexity parameter, four values are tested (5, 10, 30, and 50) instead of relying on a single visualization. For low perplexity values (5 and 10), the projection is

fragmented into small local groups, which is expected: t-SNE is paying attention only to very close neighbors. For higher values (30 and 50), the structure becomes smoother and more global.

Across all perplexity values, the benign/malignant separation is reasonably stable. One side of the plot consistently contains mostly benign samples and the other side mostly malignant ones. This stability is more convincing than any single plot, because it shows that the structure is not an artifact of a specific hyperparameter choice. Perplexity = 30 is used as the main reference visualization, because it gives a balanced view between local fragmentation and global smoothing.

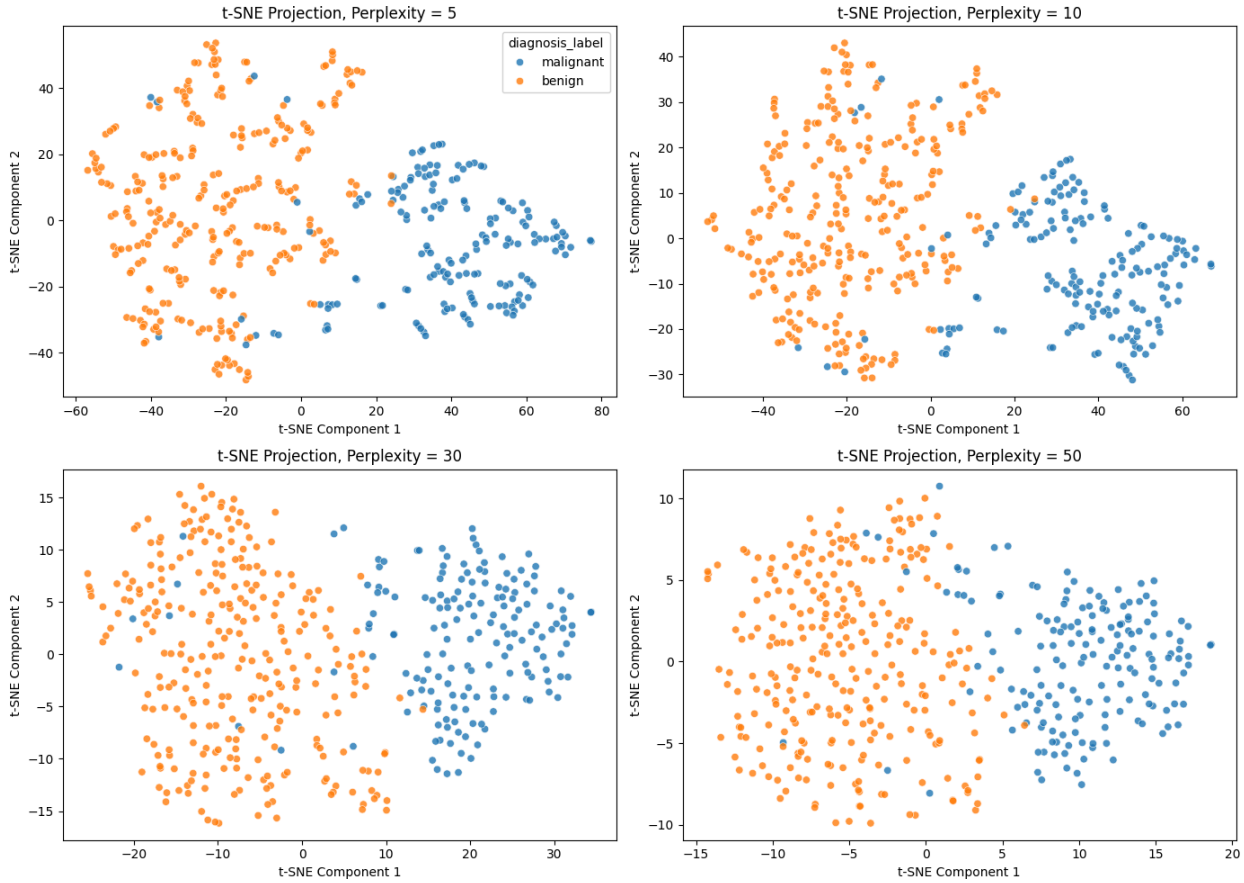


Figure 8: t-SNE projections at perplexities 5, 10, 30, and 50

3.3 Clustering

After dimensionality reduction, two clustering methods are applied to the standardized training set to check whether the data naturally forms groups. Both methods are fitted with two clusters, matching the number of known diagnostic classes, and neither uses the diagnosis labels during fitting.

K-Means.

K-Means partitions the data by assigning each point to the nearest cluster center and iteratively updating those centers. With n_clusters = 2, the algorithm finds one cluster containing mostly benign samples (270 benign, 27 malignant) and another containing mostly malignant samples (15 benign, 143 malignant). The alignment with the true diagnosis is far from accidental, even though the algorithm never saw the labels.

To measure this quantitatively, we use two complementary metrics. The Adjusted Rand Index (ARI), which compares the clusters to the true labels, is 0.662. The Silhouette Score, which measures how compact and well-separated the clusters are geometrically (without using labels), is 0.343. These two metrics answer different questions: ARI asks whether the clusters agree with the diagnosis, and the silhouette score asks whether the clusters are internally consistent.

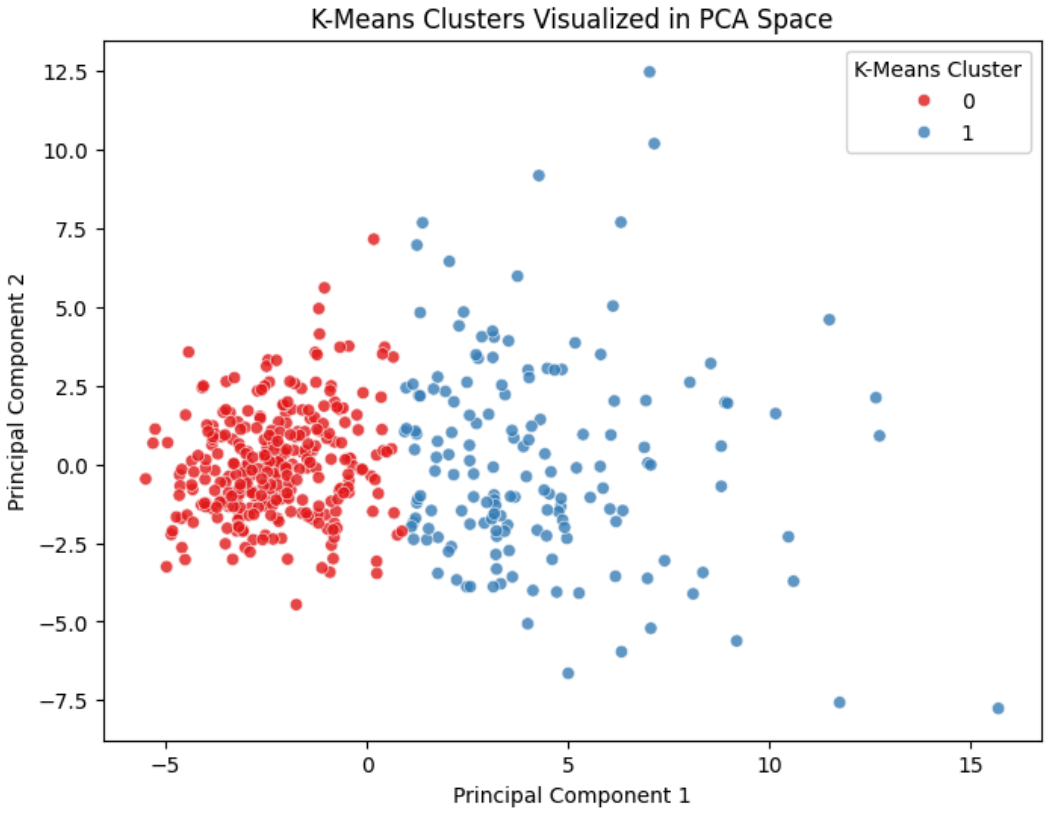


Figure 9: K-Means clusters visualized in PCA space

Gaussian Mixture Model.

A Gaussian Mixture Model with two components and a full covariance matrix assumes that the data is generated by two Gaussian distributions with possibly different shapes. This is more flexible than K-Means, which implicitly assumes spherical clusters. With the same setup as before, GMM achieves an ARI of 0.752 and a silhouette score of 0.314.

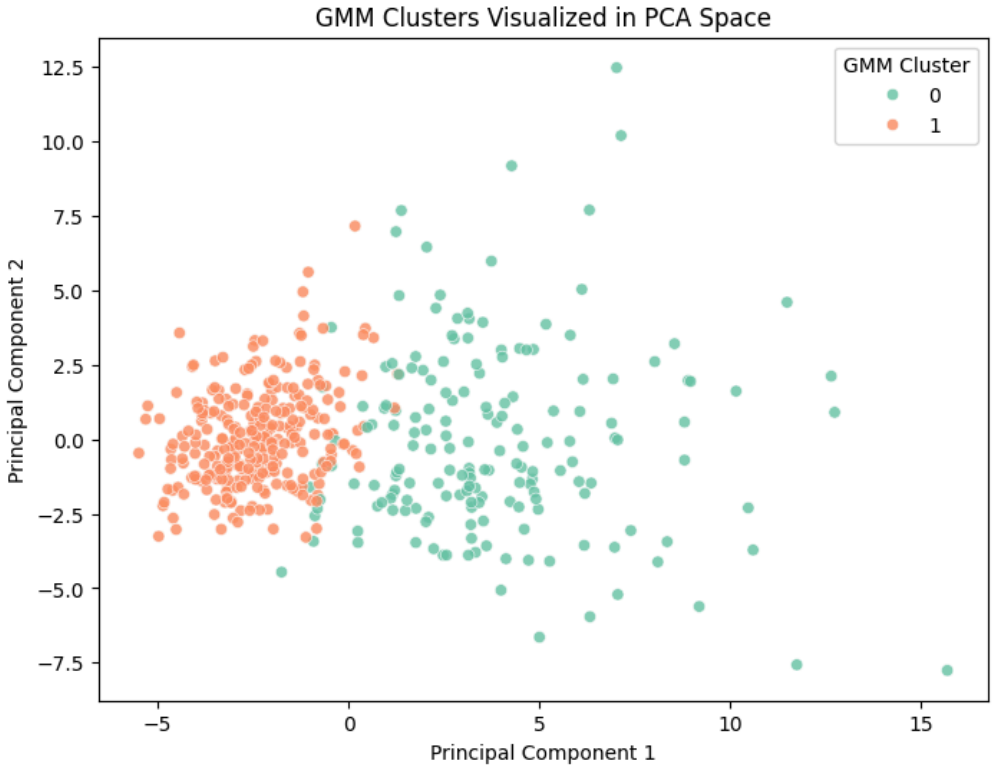


Figure 10: GMM clusters visualized in PCA space

Comparing the two methods. GMM agrees more strongly with the true diagnosis labels than K-Means (ARI 0.752 vs 0.662).On the other hand, K-Means produces clusters that are slightly more compact in the geometric sense (silhouette 0.343 vs 0.314). These two observations are not contradictory: GMM can fit elongated, elliptical clusters and therefore recovers the elongated benign/malignant shape better, while K-Means insists on round clusters and so achieves a higher silhouette by being more geometrically rigid.

Visualizing both clusterings in PCA space confirms this: both methods split the data mainly along PC1, mirroring the direction of the benign/malignant separation observed in the PCA plot. The overlap in the central region is exactly where the misclassified points lie.

The main limitation of this part of the analysis is that it forces the number of clusters to two. The dataset may contain finer substructure, such as different tumor subtypes, that two clusters

cannot capture. Both methods are also sensitive to scaling, which is why standardization was applied first.

3.4 What we learned from unsupervised learning

Across PCA, t-SNE, and the two clustering methods, the same pattern keeps appearing: the diagnostic distinction is reflected in the geometry of the feature space, mainly along the direction captured by PC1. This is a strong result on its own, because it was obtained without ever using the diagnosis labels. It also sets up the supervised part of the project: if the structure is already visible without labels, supervised models should be able to exploit it efficiently.

4: Supervised learning

After confirming that the diagnostic signal is geometrically present in the data, we now build supervised classifiers to actually predict the diagnosis. We compare three models: Logistic Regression as the basic linear baseline, Support Vector Machine with an RBF kernel as a more flexible nonlinear model, and K-Nearest Neighbors as a distance-based alternative. Each model is trained twice, once on the original 30 features and once with PCA inside the pipeline. This direct comparison is what connects the supervised and unsupervised parts of the project.

4.1 Setup

All preprocessing is included inside scikit-learn Pipelines so that scaling, and PCA where applicable, is fitted only on the training portion of each cross-validation fold. The cross-validation strategy is stratified 5-fold on the training set, which keeps the class proportions consistent across folds.
The main scoring metric for model selection is the F1 score. Because malignant is encoded as the positive class, F1 balances precision and recall for the medically critical class. Recall is also inspected carefully, because missing a malignant case is a more serious error than incorrectly flagging a benign case.

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)
scoring = "f1"
```

For each model, we use GridSearchCV with the chosen scoring metric and CV strategy. The grid for each model is small, but covers a reasonable range of regularization and complexity values.

4.2 Models on the original 30 features

The three models are first trained on the standardized original features, without any dimensionality reduction.

Logistic Regression.

A linear model that outputs a probability through a sigmoid function and learns by minimizing binary cross-entropy. The regularization parameter C is tuned over [0.01, 0.1, 1, 10, 100]. The best value is C = 1, with a cross-validation F1 score of 0.9640. The fact that a moderate amount of regularization works best is consistent with the PCA result: the data is largely linearly separable, so a simple model does not need strong shrinkage to generalize well.

Support Vector Machine (RBF kernel).

SVM with an RBF kernel allows nonlinear decision boundaries. Two hyperparameters are tuned: C in [0.1, 1, 10, 100] and gamma in ["scale", 0.01, 0.1, 1]. The best combination is C = 10 and gamma = "scale", with a cross-validation F1 of 0.9674. This is slightly better than Logistic Regression, but the improvement is small, suggesting the dataset is already very close to linearly separable in this representation.

K-Nearest Neighbors.

kNN classifies each point based on the majority class among its k closest training neighbors. The number of neighbors is tuned in [3, 5, 7, 9, 11, 15] and the weighting in ["uniform", "distance"]. The best configuration uses k = 3 with uniform weights, giving a cross-validation F1 of 0.9633.

All three models perform very similarly on the original features, with F1 scores between 0.9633 and 0.9674. This narrow spread reinforces the earlier conclusion that the dataset is geometrically clean and the diagnostic signal is strong.

4.3 Models with PCA inside the pipeline

The same three models are then trained with PCA included as a step inside the pipeline. The number of components is tuned over [2, 6, 7, 10, 15]. The values 7 and 10 are included because they retain approximately 90% and 95% of the variance respectively, while 2 connects to the PCA visualization and 6 and 15 cover slightly more compressed or more relaxed alternatives.

The structure of the pipeline is identical across models:

```
svm_pca_pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("pca", PCA()),
    ("model", SVC(kernel="rbf", probability=True, random_state=RANDOM_STATE))
])
```

Because the PCA step lives inside the pipeline, it is fitted only on the training folds during cross-validation. This is critical for avoiding any data leakage.
The results, ranked by cross-validation F1, are:

- SVM RBF + PCA: F1 = 0.9698 (PCA = 15, C = 10, gamma = 0.01)
- Logistic Regression + PCA: F1 = 0.9671 (PCA = 10, C = 1)
- kNN + PCA: F1 = 0.9640 (PCA = 6, k = 3, uniform weights)

Compared with the original-feature versions, every model either matches or slightly improves once PCA is added. The largest improvement comes from SVM, which goes from 0.9674 to 0.9698 with PCA = 15 and a smaller gamma. This is a small but consistent gain, and it is exactly the kind of effect the unsupervised analysis predicted: by collapsing the strong correlations between redundant features, PCA gives the supervised models a slightly cleaner input space to work with.

4.4 Model selection

The final model selected for evaluation is the SVM with RBF kernel and PCA inside the pipeline, with C = 10, gamma = 0.01, and 15 components. Its cross-validation F1 score of 0.9698 is the best among all six configurations. Logistic Regression with PCA is essentially tied with it (0.9671), so in a real medical setting the interpretability advantage of Logistic Regression might tip the balance the other way. For the purposes of this project, we follow the cross-validation result and proceed with the SVM model.

5: Test-set evaluation

The selected model is now evaluated on the held-out test set for the first and only time. Because all preprocessing and hyperparameter tuning was performed on the training set, this evaluation gives an honest estimate of how the model would behave on new patients.

5.1 Final metrics

The test set contains 114 samples: 72 benign and 42 malignant. The final metrics for the SVM + PCA model are:

- Accuracy: 0.9825
- Precision: 1.000
- Recall: 0.9524
- F1 score: 0.9756
- ROC AUC: 0.9967

The model classifies almost all test samples correctly, and the ROC AUC of nearly 1 confirms that it ranks malignant cases very well, regardless of the chosen decision threshold.

5.2 Classification report

Breaking the metrics down by class:

- Benign: precision = 0.97, recall = 1.00, F1 = 0.99, support = 72
- Malignant: precision = 1.00, recall = 0.95, F1 = 0.98, support = 42

Every benign case is correctly identified, and no benign case is incorrectly flagged as malignant. Two malignant cases out of 42 are missed by the model.

5.3 Confusion matrix

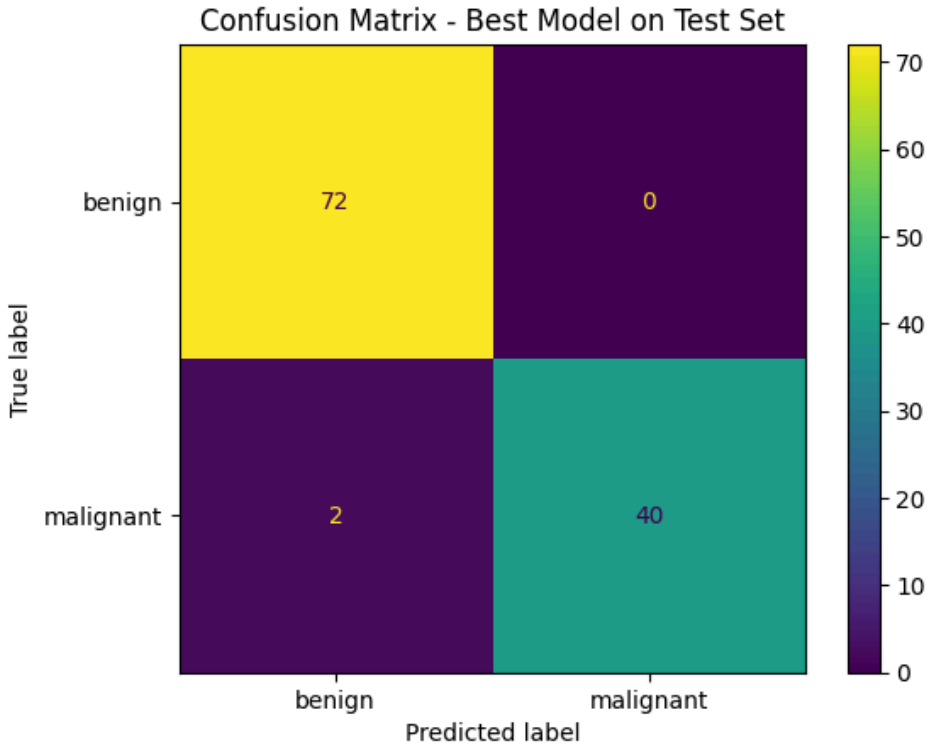


Figure 11: Confusion matrix on the test set

The confusion matrix shows zero false positives (no benign samples labeled malignant) and two false negatives (two malignant samples labeled benign).

5.4 Medical interpretation

In a medical screening context, the two types of error are not symmetric. A false positive means a benign tumor is wrongly flagged as malignant; this is unpleasant for the patient but ultimately leads to further testing, which can correct the mistake. A false negative means a malignant tumor is wrongly labeled as benign, and that is the much more serious error, because it can delay treatment.

With this in mind, the model's profile is interesting. Precision is perfect, so the model never raises a false alarm. Recall is high but not perfect at 0.95, which means about one in twenty malignant cases is missed. In a real clinical pipeline, this would be the dial to adjust: lowering the decision threshold would push recall up at the cost of some precision. Because the model also outputs probabilities (the SVM was trained with probability = True), this kind of threshold tuning is straightforward.
The ROC AUC of 0.9967 indicates that the model ranks malignant cases extremely well; almost any reasonable threshold gives strong performance.

6: Connecting unsupervised and supervised learning

One of the explicit goals of the project is to connect the unsupervised and supervised parts of the analysis. The choice of methods makes this connection direct, since the same PCA step appears in both halves of the project. Four points stand out.

6.1 PCA improves prediction marginally but consistently

Every model with PCA in the pipeline matches or slightly improves on its original-feature counterpart. The best configuration overall (SVM + PCA, F1 = 0.9698) uses PCA as a preprocessing step. The improvement is small but consistent across all three model families, which suggests that it is not an accident.
The reason traces directly back to Section 2.4: the correlation matrix showed that several features carry almost identical information (radius, perimeter, and area in particular). PCA collapses these correlated groups into orthogonal components, which gives margin-based and distance-based models a slightly cleaner input space.

6.2 The clusters foreshadow the classifier

The unsupervised clustering already pointed toward a strong supervised result. GMM achieved an ARI of 0.752 without ever seeing the labels, and K-Means achieved 0.662. An ARI in that range means the unlabeled clusters already capture most of the diagnostic distinction, which sets a high ceiling for what a supervised model can do on top.

The supervised models then reach an F1 close to 0.97 on cross-validation and 0.98 on the test set, which is consistent with that prediction. The supervised step is mostly refining a boundary that was already roughly visible in the unlabeled data.

6.3 Same axis throughout

A more qualitative observation: across PCA, K-Means, GMM, and the supervised models, the same direction keeps coming up. The first principal component is where benign and malignant separate in the PCA plot, it is the axis along which both clustering methods split the data, and after PCA is included in the supervised pipeline, SVM and Logistic Regression effectively learn

decision boundaries that live in this same rotated space. The fact that several independent methods converge on the same axis is strong evidence that the structure is real and not a methodological artifact.

6.4 Possible extensions

One natural extension would be to cross-reference the misclassified test points with the cluster assignments. If the two malignant cases that the model misses fall in the overlap region between clusters in PCA space, that would confirm that the supervised error and the unsupervised ambiguity have the same source. This is left as a possible follow-up.

7: Reflection

This final section briefly reflects on what worked, what did not, what surprised us, which method felt most convincing, and what the limitations of the study are.

What worked well. The pipeline-based design worked very smoothly. Putting scaling and PCA inside sklearn Pipelines made the cross-validation results trustworthy, and the GridSearchCV approach made the model comparison fair. On the unsupervised side, GMM clearly beat K-Means at recovering the benign/malignant structure, which was a satisfying example of how a more flexible distributional assumption can outperform a simpler geometric one when the clusters are not round.

What did not work as well. Forcing the number of clusters to two was reductive. The dataset almost certainly contains finer substructure (different tumor subtypes, different stages), and a more careful exploration with three or four clusters might have revealed it. t-SNE was also harder to evaluate than expected: the visualizations are striking, but there is no clean quantitative way to score them.

What surprised us. How small the gap between Logistic Regression and the more flexible models turned out to be. Going in, we expected SVM with an RBF kernel to clearly beat the linear baseline. In practice, the dataset is already so close to linearly separable after standardization that the linear model leaves very little room for improvement, even for a nonlinear competitor.

Which method was most convincing. Strictly on cross-validation F1, SVM with PCA wins. But Logistic Regression with PCA is essentially tied with it and has a clear interpretability advantage: each coefficient corresponds to a principal component, and the loadings of that component on the original features can be inspected. In a real medical context, that traceability would probably make Logistic Regression the more defensible choice.

Limitations. The dataset is small (569 samples) and represents a single source. The 30 features are hand-engineered summaries of cell-nucleus images rather than raw pixel data, which limits how directly the conclusions transfer to modern image-based pipelines. We also used F1 as a single selection criterion, which weighs precision and recall equally; in a clinical

setting, recall (sensitivity) would typically be weighted more heavily because of the asymmetry between false positives and false negatives discussed in Section 5.4.

8: Conclusion

The Breast Cancer Wisconsin dataset has a geometric diagnostic signal. PCA shows that this signal is mostly along the first principal component, t-SNE confirms it across multiple perplexities, and both K-Means and Gaussian Mixture Models recover most of the benign/malignant distinction without using labels. On the supervised side, three different model families all reach cross-validation F1 scores between 0.96 and 0.97, and the best configuration (SVM with RBF kernel and PCA inside the pipeline) achieves 98.25% accuracy, perfect precision, 95.24% recall, and a ROC AUC of 0.9967 on the test set. The unsupervised and supervised parts of the project agree on the same underlying structure, which is the cleanest possible outcome: the geometry of the input features already encodes the diagnosis, and the supervised model simply learns to draw the right boundary in that space.